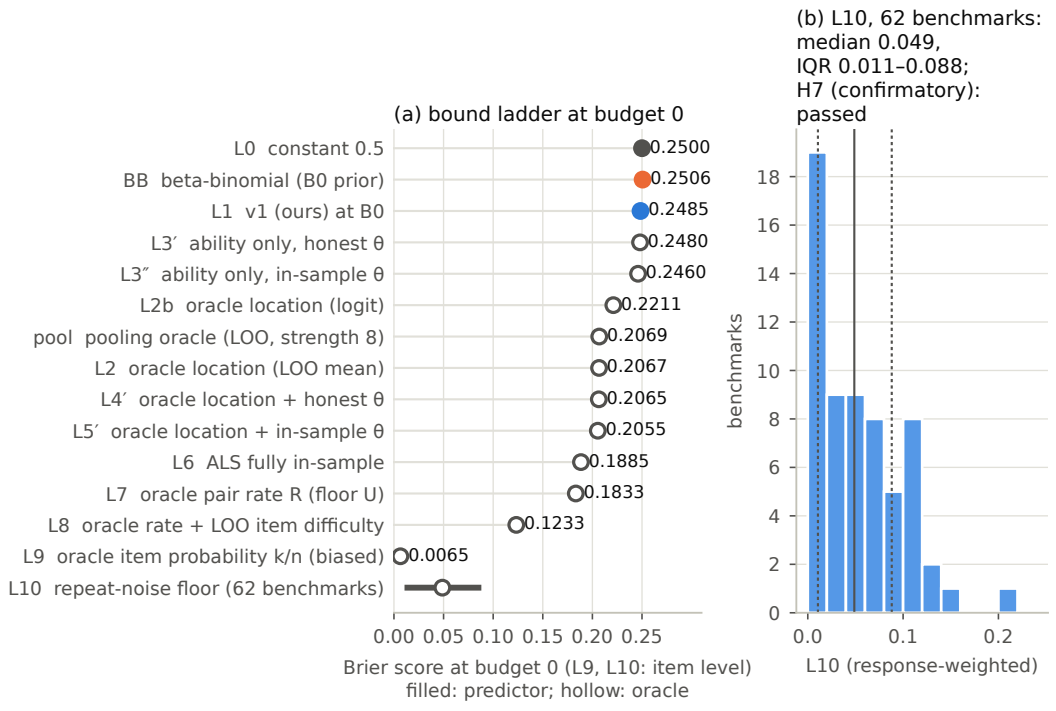
(a) bound ladder at budget 0
L0 constant 0.5
BB beta-binomial (B0 prior)
L1 v1 (ours) at B0
L3′ ability only, honest θ
L3″ ability only, in-sample θ
L2b oracle location (logit)
pool pooling oracle (LOO, strength 8)
L2 oracle location (LOO mean)
L4′ oracle location + honest θ
L5′ oracle location + in-sample θ
L6 ALS fully in-sample
L7 oracle pair rate R (floor U)
L8 oracle rate + LOO item difficulty
L9 oracle item probability k/n (biased)
L10 repeat-noise floor (62 benchmarks)
0.2500
0.2506
0.2485
0.2480
0.2460
0.2211
0.2069
0.2067
0.2065
0.2055
0.1885
0.1833
0.1233
0.0065
0.00 0.05 0.10 0.15 0.20 0.25
Brier score at budget 0 (L9, L10: item level)
filled: predictor; hollow: oracle
(b) L10, 62 benchmarks:
median 0.049,
IQR 0.011–0.088;
H7 (confirmatory):
passed
benchmarks
0 2 4 6 8 10 12 14 16 18
0.0 0.1 0.2
L10 (response-weighted)